DBMS PROJECT REPORT

UniChrist: Integrated Campus Life & Location Services Portal

Submitted By

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Project Guide

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A project report submitted in partial fulfillment of the requirements for the course BCA381-4 – DBMS Project of the degree of Bachelor of Computer Applications (BCA) of CHRIST (Deemed to be University).

Course: BCA381-4 — DBMS Project | Class: 4BCA-B
Academic Year: 2025 – 2029

DBMS Project Specification Report

Problem Statement, Domain Analysis & System Requirements Specification

Section 1: Team Details

Course: BCA381-4 — DBMS Project | Class: 4BCA-B | Project Guide: Dr Hemanth K S

S.No	Student Name	Register Number	Role in Project
1	Mayank Raj Gupta	2543115	Team Lead: Database design, table schema, SQL scripts and backend logic
2	Sonti Sri Lakshmi Aslesha	2543140	Schema Normalization (1NF to BCNF), constraints, views and queries
3	Aadiya Naresh	2543163	System analysis, UI flow, test cases and documentation

Section 2: Chosen Application Domain & Problem Statement

Chosen Domain: Smart Campus Management & Schedule-Based Location Services System

Why we chose this project: As students at the Bangalore Yeshwanthpur campus, our team experiences everyday friction because different campus facilities run on separate, unlinked methods. Specifically:

- **Finding Teachers:** Students spend 15 to 25 minutes walking across Block A and Block B staff rooms trying to find a teacher during free hours, only to realize the teacher is currently taking a lecture in a classroom, conducting a practical lab, or attending a meeting. Live sensor tracking is impractical and expensive across an entire campus.
- **Cafeteria Lines:** During our 20-minute morning and lunch breaks, students crowd food counters just to place an order, leaving hardly any time to eat.
- **Scattered Services:** The bookstore, uniform desk, campus health room, lost & found counter, and parking spots all keep separate logbooks with no live lookup.
- **Maintenance Complaints:** When Wi-Fi drops in lab workstations or classroom projectors malfunction, complaints written in physical logbooks get delayed or forgotten with no status update.
- **Finding Rooms:** New students and visitors frequently get confused trying to find specific labs, seminar halls, and faculty cabins across floors.

Section 3: Business & Application Objectives

Our project aims to achieve the following 6 practical objectives:

1. **Centralized Database:** Design and build a clean relational database normalized up to BCNF that brings 12 essential campus services together without duplicate records.
2. **Automated Timetable-Based Location Engine:** Build an intelligent schedule-matching database engine that checks the faculty timetable against the current clock: showing the classroom/lab during lecture hours, or the staff room during free hours.
3. **Online Cafeteria Pre-Ordering:** Allow students to view the live canteen menu, place food orders ahead of time, get a digital token number, and automatically reduce available kitchen stock.
4. **Campus Utility Tracking:** Digitize the health room visit logs, lost-and-found claims, and block-wise vehicle parking slot counts.
5. **Trackable Grievance Ticketing:** Build a complaint portal where student issues follow a clear status lifecycle (Pending -> Assigned -> In-Progress -> Resolved) with timestamps.
6. **Multi-Floor Room Directory:** Provide a simple floor-by-floor campus map where users click a block and floor number to find any classroom, lab, or seminar hall.

Section 4: Detailed Module Breakdown (All 12 Modules)

The application covers 12 modules, each supported by dedicated relational database tables:

Module No.	Campus Service	Main Database Tables	What This Module Does
1	Cafeteria Ordering	Cafeteria_Vendors, Items, Orders, Order_Items	Menu browsing, pre-ordering, token generation, kitchen queue, live stock update.
2	Bookstore & Store	Store_Categories, Store_Inventory, Store_Orders	Checks textbook & uniform stock, allows booking, creates digital receipt.
3	Medical Room	Medical_Visits, Medical_Prescriptions	Logs student check-ins, duty doctor hours, first-aid alerts, medicine records.
4	Lost & Found	Lost_Found_Items, Lost_Found_Claims	Uploads photos of found items, collects claims, logs security handovers.
5	Library OPAC	Book_Categories, Library_Books, Circulation	Book searches, issue/return dates, reading desk bookings, overdue fine calculation.

Module No.	Campus Service	Main Database Tables	What This Module Does
6	Smart Parking	Parking_Slots, Parking_Logs	Shows free 2-wheeler, 4-wheeler, and EV charging slots; logs security gate entry.
7	Campus Events	Campus_Events, Event_Registrations	Central calendar for college fests and workshops, student signups, digital passes.
8	College Clubs	Student_Clubs, Club_Memberships	Lists clubs under Sciences, Commerce, Arts & Management; manages member lists.
9	Teacher Schedule & Location	Faculty_Timetables, Faculties, Campus_Rooms	Fetches period schedule: shows current class room; if free, shows staff room.
10	Grievance Portal	Grievances, Grievance_Audit_Logs	Students log Wi-Fi, lab, or classroom issues; tracks fix status with staff notes.
11	Study Resources	Academic_Resources	Searchable repository for syllabus copies, past exam question papers (PYQs), lab manuals.
12	Campus Map	Campus_Blocks, Campus_Rooms	Interactive directory: pick block and floor to see all classrooms, labs & halls.

Section 5: Existing System vs. Proposed UniChrist System

A detailed comparison between how work is currently done on campus and our database solution:

Campus Service	How It Is Done Right Now	Problems with Current Method	How UniChrist Fixes It
Finding Faculty	Walking to staff rooms, asking classmates or peons	Wasted time, teachers often in another class or meeting	Checks faculty timetable: shows active classroom during lectures, or staff room if free.
Canteen Orders	Standing in long queues, paying at counter, paper slips	Crowded counters, 20-min breaks wasted, food items run out	Pre-order on phone, get token number, collect food when ready.
Lost & Found	Paper book kept at the main security guard post	Students don't know items were found; items sit unclaimed	Students browse photos of found items online and verify ownership.
Complaints	Handwritten register in maintenance office or verbal complaints	Complaints get lost, no record of who is fixing it or when	Digital ticket with timestamp; shows Pending, In-Progress or Resolved.
Campus Directions	Looking at wall signboards or asking security guards	Hard to locate specific computer labs and seminar halls across floors	Clean map interface: select Block -> Floor to see exact room numbers.

Section 6: Feasibility Study

We analyzed whether our project is realistic and achievable across four key areas:

Feasibility Area	What We Examined	Our Findings & Project Reality
Technical	Database software, programming tools, hardware requirements	FEASIBLE: Timetable lookup is much more reliable and feasible than expensive biometric hardware. The database queries Faculty_Timetables with current day and time to return the room in milliseconds. Built with PostgreSQL / SQL Server and .NET.
Operational	Ease of use for students, teachers, cafeteria staff and security	FEASIBLE: All students and teachers already carry smartphones. The screens are simple and easy to understand. Each user type (student, teacher, canteen vendor, security guard) only sees options relevant to them.
Economic	Development costs, software licenses, operational expenses	FEASIBLE: By using the database timetable engine instead of installing physical RFID/biometric readers at every door, hardware costs are eliminated. We use free, open-source DBMS tools available in university labs.
Behavioural	Teacher privacy, student willingness, campus regulations	FEASIBLE: Uses only official university timetable data already published by departments. Zero privacy concerns because it reflects public academic schedules, not personal physical tracking.

Section 7: Software Requirement Specifications — Functional Requirements

These functional requirements describe what the system must do across all modules:

Req ID	Priority	Module	System Action & What It Does
FR-01	High	Login & Roles	Verify university user credentials and grant access according to role (Student, Faculty, Vendor, Medical, Admin).
FR-02	High	Timetable Storage	Store weekly period schedules for all faculty members (day of week, start time, end time, room ID, subject code, class section).
FR-03	High	Teacher Location	Fetch current time and match with teacher timetable: if in class, show classroom/lab; if free, show designated staff room.
FR-04	High	Canteen Order	Allow students to view food items, place an order, get an order token, and automatically reduce item stock.
FR-05	Medium	College Store	Keep count of textbooks, uniform items, and stationery; let students reserve available items online.
FR-06	High	Health Room	Log student medical visits, record symptoms and first-aid given, and send emergency alerts if needed.
FR-07	Medium	Lost & Found	Let users post pictures and descriptions of found items; allow students to submit verification claims.
FR-08	High	Library Search	Search books by title or author, track borrowed books, calculate overdue fines, and book reading room desks.
FR-09	Medium	Parking Slots	Show counts of available 2-wheeler, 4-wheeler, and EV charging parking spots across campus blocks.
FR-10	Medium	Events Calendar	Display upcoming fests and guest lectures, register students, and generate digital entry passes.
FR-11	Medium	Clubs Module	Show club details under Sciences, Commerce, Arts, and Business; handle student member signups.
FR-12	High	Grievance Ticket	Allow students to lodge maintenance or Wi-Fi complaints and see status move from Pending to Resolved.

Req ID	Priority	Module	System Action & What It Does
FR-13	Medium	Study Material	Provide downloadable PDF syllabus copies, past exam question papers (PYQs), and practical lab manuals.
FR-14	High	Campus Map	Display an interactive directory where clicking a block and floor reveals all classrooms, labs, and halls.
FR-15	Medium	Admin Reports	Generate summary reports for daily canteen sales, active maintenance complaints, and library book returns.
FR-16	High	Notifications	Send alert notifications when food is ready for pickup, a complaint is updated, or a borrowed book is due.

Section 8: Non-Functional Requirements

These requirements define quality, speed, security, and database reliability standards:

Req ID	Quality Factor	How Our Database & Application Handles It
NFR-01	Response Time	Search queries for teacher schedule lookup, canteen menus, and library books must respond in under 500 milliseconds. We use B-Tree indexes on faculty_id and day_of_week.
NFR-02	Security	Student and faculty passwords are encrypted using Bcrypt. SQL injection is prevented by using parameterized queries and stored procedures.
NFR-03	Data Consistency	All orders and book borrowing transactions follow strict ACID rules. We use database row locks (SELECT FOR UPDATE) to stop double-booking.
NFR-04	Ease of Use	Interface works smoothly on mobile phones and laptops. Any core task can be completed in three clicks or taps.
NFR-05	Availability	The database is available throughout campus hours (7:30 AM to 7:30 PM) with regular daily database backup copies.
NFR-06	Clean Schema	All database tables are normalized up to BCNF to ensure there is no repetitive data or accidental deletion of records.

Section 9: Software Development Model & Guide Discussion

Development Model Chosen: Agile / Iterative Model

Why this model fits: Because we are building 12 modules, doing everything in one giant step is risky. The Agile iterative approach lets our team build, test, and normalize the database tables in two-week cycles (Sprints):

- **Sprint 1 (Weeks 1-2):** User accounts, role access, campus blocks, rooms, and teacher schedule timetable engine.
- **Sprint 2 (Weeks 3-4):** Canteen menu, pre-ordering, bookstore inventory, and medical room visit logs.
- **Sprint 3 (Weeks 5-6):** Library OPAC, parking slot counters, lost & found, and student clubs/events.
- **Sprint 4 (Weeks 7-8):** Grievance redressal ticketing, academic resources, and campus floor map directory.
- **Sprint 5 (Weeks 9-10):** Stored procedures, triggers, views, load testing, and final project documentation.

Guide Discussion Notes (Meeting with Dr Hemanth K S):

No.	Guide Advice & Suggestion	How We Are Implementing It in Our Project
1	Feasibility of Teacher Tracking	Dr Hemanth K S advised that continuous live hardware/biometric tracking across campus is difficult to implement. He instructed us to fetch the teacher's schedule from the database: if they have a class, show the classroom/lab; if free, show their designated staff room.
2	Normalize all tables up to BCNF	Split user and faculty records properly; use junction tables for student club memberships and fest signups.
3	Prevent food stock overselling	Use database transaction row locks in stored procedure sp_PlaceCafeteriaOrder so stock never goes below zero.
4	Lock grievance status transitions	Use a database trigger to enforce that complaints move in order: Pending -> In-Progress -> Resolved.

Section 10: Student Declaration & Guide Verification

We confirm that the information provided in this document is based on our project discussion and preliminary analysis. We have explained the proposed project to our guide and discussed the scope and requirements.

Team Members Confirmation:

- **Member 1 (Team Lead):** Mayank Raj Gupta (Register No: 2543115) — Signature: _____
- **Member 2:** Sonti Sri Lakshmi Aslesha (Register No: 2543140) — Signature: _____
- **Member 3:** Aadiya Naresh (Register No: 2543163) — Signature: _____

Date of Submission: _____

Project Guide Verification & Remarks:

Project Guide Name: Dr Hemanth K S

Department: Department of Computer Science, CHRIST (Deemed to be University)

Guide's Signature: _____ Date: _____

Guide's Decision: [ ] Approved as submitted [ ] Approved with minor modifications [ ] Needs revision

Guide's Detailed Feedback & Suggestions:

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Suggested Assessment Rubric (For Faculty Use — Max 10 Marks):

Assessment Criterion	Max Marks	Marks Awarded
Problem statement, business objectives and comprehensive project scope	2.0	
Existing-system analysis, preliminary investigation and identification of need	2.0	
Feasibility study (Technical, Operational, Economic, Behavioural)	2.0	
Functional and Non-functional requirements / SRS completeness & clarity	2.0	
Guide discussion, technical explanation, completeness and signed verification	2.0	
TOTAL ASSESSMENT-1 MARKS	10.0	

Section 11: Relational Database Architecture Blueprint (Appendix)

Summary of all 18 core relational database tables designed for UniChrist, featuring the timetable-based location engine:

Table Name	Primary Key	Foreign Keys	Key Fields & Constraints
Users	user_id (PK)	role_id -> Roles	register_number (UQ), full_name, email (UQ), password_hash, is_active
Faculties	faculty_id (PK)	user_id, dept_id, staff_room_id	designation, office_extension, consultation_hours, default_cabin
Faculty_Timetables	schedule_id (PK)	faculty_id, room_id	day_of_week, period_no, start_time, end_time, subject_code, section
Faculty_Status_Overrides	override_id (PK)	faculty_id	override_date, start_time, end_time, status (MEETING, ON_LEAVE), notes
Campus_Blocks	block_id (PK)	None	block_code (UQ), block_name, total_floors, latitude, longitude
Campus_Rooms	room_id (PK)	block_id -> Campus_Blocks	room_number, floor_number, room_type, seating_capacity (UQ block, room)
Cafeteria_Items	item_id (PK)	vendor_id -> Vendors	item_name, category, price, is_available, current_stock_qty
Cafeteria_Orders	order_id (PK)	user_id, vendor_id	order_timestamp, token_number, total_amount, order_status
Store_Inventory	product_id (PK)	category_id -> Store_Categories	product_code (UQ), product_name, unit_price, stock_quantity
Medical_Visits	visit_id (PK)	student_id, attending_staff_id	checkin_time, symptoms_observed, is_emergency_flag, disposition
Lost_Found_Items	item_id (PK)	reported_by_user_id, room_id	item_title, category, photo_url, date_found, status
Library_Books	book_id (PK)	category_id -> Book_Categories	isbn (UQ), title, author, edition, total_copies, available_copies

Table Name	Primary Key	Foreign Keys	Key Fields & Constraints
Library_Circulation	issue_id (PK)	book_id, user_id	issued_date, due_date, returned_date, fine_amount, circulation_status
Parking_Slots	slot_id (PK)	block_id -> Campus_Blocks	slot_code (UQ), vehicle_type, is_ev_charging, is_occupied
Campus_Events	event_id (PK)	club_id, venue_room_id	event_title, event_type, start_time, end_time, max_registrations
Student_Clubs	club_id (PK)	school_id, faculty_coordinator_id	club_name (UQ), club_tagline, founded_year
Grievances	ticket_id (PK)	student_id, room_id, assigned_to	category, ticket_subject, description, priority, status
Academic_Resources	resource_id (PK)	dept_id, uploaded_by_faculty	resource_type, course_code, document_title, file_url